Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
A		**2600070**

Curing Battery Anxiety: A Physics-Informed Stochastic Model for Precise TTE Prediction and System Optimization

Summary

The battery lifespan of smartphones is affected by the **nonlinear coupling** of hardware power consumption, internal resistance, and ambient temperature, which invalidates traditional linear estimation methods and triggers users' "range anxiety". As battery discharge is a dynamic process involving **electrochemical polarization**, pure data fitting fails to reveal its intrinsic mechanism. Thus, there is an urgent need to construct a continuous-time model based on **circuit dynamics** for accurate Time-to-Empty (TTE) prediction and energy management optimization.

Problem 1: Physics-Informed Continuous Discharge Modeling. To simulate the temporal evolution of battery states, we developed a physics-informed continuous model based on the **Thevenin Equivalent Circuit**. This model employs a first-order **RC differential equation** to characterize the **polarization effect** and **transient response**, and integrates a mechanism-driven component-level power model that aggregates dynamic loads of the screen, **CPU**, and network modules into a time-varying current input. Numerical solutions reveal the quasi-linear SOC decay trajectory over time and verify the linear mapping between discharge rate and average load current.

Problem 2: Prediction, Uncertainty Quantification, and Attribution. First, we calculated deterministic runtimes under different activities via **numerical integration**. Second, to address real-world variability, we implemented **Monte Carlo simulation** to generate TTE's probability distribution and quantify the tail risk of unexpected shutdowns. Third, **Global Sensitivity Analysis (Sobol Indices)** was used for power consumption attribution, identifying screen brightness and CPU load as dominant drivers, while revealing that weak signal strength causes exponential power consumption growth. Finally, we derived the **optimal control strategy** to maximize runtime under performance constraints based on these sensitivities.

Problem 3: Model Robustness Verification and Sensitivity Analysis. By relaxing idealized assumptions, we rigorously evaluated the model's reliability for **electro-thermal coupling** and **voltage cutoff paradox**. Using the **normalized local sensitivity index**, we quantified parameter impacts, revealing how aging-induced nonlinear effects and the **Peukert effect** alter discharge efficiency. Additionally, dynamic load fluctuation simulations prove the model can capture the **battery recovery phenomenon** during mode switching (intermittent loads), verifying its effectiveness in complex time-varying usage scenarios.

Finally, we summarized the model's advantages and disadvantages, discussed its potential applications, proposed improvements to manufacturers for the operating system's **energy scheduling strategy**, and attached a battery optimization guide for smartphone users.

Keywords: Thevenin Battery Model; Continuous-Time Modeling; Remaining Runtime Prediction; Global Sensitivity Analysis; Pareto Optimality

Contents

1 Introduction

1.1 Background

Lithium-ion batteries serve as the heartbeat of modern smartphones, powering an array of components from high-resolution screens to multi-core processors. Despite advances in battery density, the estimation of remaining battery lifeoften represented as State of Charge (SOC) or Time-to-Emptyremains a complex challenge. Users frequently experience "battery anxiety" when the displayed percentage does not linearly correspond to usage time, particularly when the phone shuts down unexpectedly under heavy loads (e.g., gaming or navigation) despite showing positive charge. This discrepancy arises because battery depletion is not merely a subtraction of energy; it is a dynamic physicochemical process influenced by internal resistance, polarization effects, and voltage constraints. Therefore, developing a mathematical model that accounts for these internal dynamics is essential for accurate prediction.

1.2 Literature Review

State-of-charge (SOC) estimation and remaining-dischargeable-time (RDT) prediction are critical for lithium-ion battery management systems (BMS) [3, 4]. Traditional Coulomb counting suffers from cumulative errors [5], while data-driven methods lack physical interpretability [6, 7]. Model-based approaches dominate high-precision scenarios: equivalent circuit models (ECMs) are efficient but limited [8, 9], and electrochemical models like the SFP model offer superior accuracy [10, 11]. Recent advances integrate Lebesgue sampling (LS) and uncertainty management to optimize computation and precision, yet balancing model fidelity, efficiency, and robustness remains a key challenge.

1.3 Restatement of the Problem

Our core task is to develop a continuous-time mathematical model for describing the State of Charge (SOC) of a smartphone lithium battery and predicting the Time-to-Empty under realistic usage scenarios. The specific requirements for each phase are as follows:

- **Problem 1 (Model Construction)**: Establish a system of equations for SOC with time as the independent variable, moving beyond simple linear depletion descriptions, incorporate power-consuming factors such as the screen, processor, and network as well as the battery's real operational characteristics, estimate parameters and validate the framework using compliant public data, and simultaneously visualize the power depletion process and primary power sources.
- **Problem 2 (Battery Life Prediction)**: Calculate the Time-to-Empty under different initial SOC levels and usage scenarios based on the model, quantify prediction uncertainty by comparing with measured or reasonable behaviors, clarify the model's applicable and ineffective scenarios, and explain the core causes of battery life differences while identifying key high/low power-consuming factors.
- **Problem 3 (Sensitivity Analysis)**: Explore the variation laws of the model's prediction results when adjusting modeling assumptions, changing parameter values, and fluctuating usage patterns.

- **Problem 4 (Optimization Recommendations)**: Combine insights from the model to propose user-side battery life optimization behaviors and operating system-level power-saving strategies, while also considering the impact of battery aging and providing adaptation ideas for extending the model to other portable devices.

1.4 Our Work

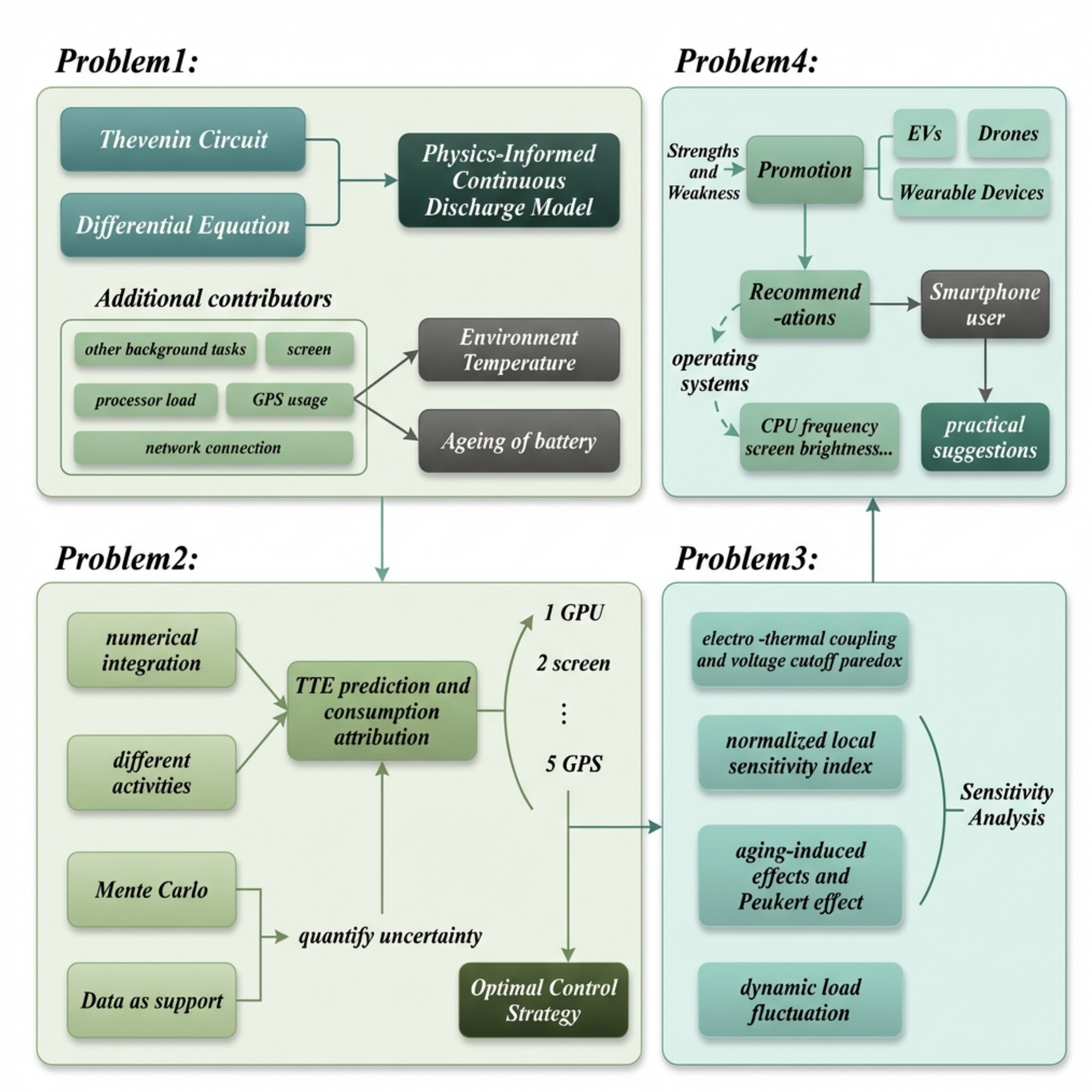

Figure 1: our work

2 Assumptions and Justifications

To render the modeling tractable while preserving physical significance, we make the following assumptions:

Assumption 1: The battery behaves as a first-order **Thevenin Equivalent Circuit**, and smartphone component power consumption is **additive** with a **continuous and monotonic** discharge process.
Justification: Despite complex electrochemical reactions, a first-order RC circuit captures key dynamics (Ohmic resistance and polarization) for sufficient SOC estimation, avoiding higher-order model complexity; total battery power draw equals the sum of individual subsystem power (Screen, CPU, GPU, etc.) plus base static power (negligible cross-coupling effects), and we assume no intermittent charging occurs during discharge (Time-to-Empty) simulation.

Assumption 2: "Empty" is defined by either SOC = 0% or **Terminal Voltage** $\leq$ **Cutoff Voltage**.
Justification: In practice, **Battery Management Systems (BMS)** enforce a voltage floor (typically 3.0V) to prevent permanent chemical damage; phones shut down when this limit is breached, even with residual theoretical charge.

Assumption 3: For the baseline model, temperature is **constant within a specific scenario**.
Justification: While temperature fluctuates, assuming an average effective temperature for specific activities (e.g., 40řC for gaming, 25řC for standby) enables clearer parameter estimation without initial complex thermal differential equations.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations Used in This Paper

Symbol	Description	Unit
t	Time variable	s or h
$\text{SOC}(t)$	State of Charge of the battery at time t (0 to 1)	Dimensionless
$V_{\text{term}}(t)$	Terminal Voltage (measured at battery poles)	V
$U_{\text{ocv}}(\text{SOC})$	Open Circuit Voltage, a non-linear function of SOC	V
$U_1(t)$	Polarization voltage across the RC network	V
$I(t)$	Discharge current (positive for discharge)	A
$P_{\text{total}}(t)$	Total power demand from smartphone components	W
Q_{total}	Total battery capacity	C or Ah
R_0	Ohmic internal resistance of the battery	Ω
R_1	Polarization resistance	Ω
C_1	Polarization capacitance	F
V_{cutoff}	Cutoff voltage threshold for device shutdown	V
T_{empty}	Predicted Time-to-Empty	h

4 Problem 1:Physics-Informed Continuous Battery Discharge Model (Model 1)

This chapter aims to establish a continuous-time mathematical model with deep integration of data-driven methods and physical mechanisms, so as to accurately predict the battery depletion time (Time-to-Empty, TTE) of smartphones under different usage scenarios. Considering that pure data fitting is difficult to explain the nonlinear behaviors under extreme conditions (e.g., voltage drop under low temperatures), we adopt a physics-load coupling modeling strategy: first, extract the battery electrochemical parameters from the discharge curves in the AndroWatts dataset to establish a dynamic physical model of the battery; then, conduct multivariate statistical analysis using the hardware state logs in the dataset to determine the power consumption coefficients of each component; finally, achieve accurate simulation of the discharge process through numerical solution.

4.1 Electrochemical Battery Dynamics with Temperature and Aging Effects

To construct the continuous-time model required by the problem and to deeply reveal the complex nonlinear relationship between battery charge and terminal voltage, we abandon the simple linear Coulomb counting method and instead establish a physical model based on the First-Order Thevenin Equivalent Circuit. The establishment of this model structure is rooted in physical principles, while its key parameters are obtained through system identification of the dataset.

First, the core of describing the evolution of the battery's remaining chemical energy is the rate equation of the State of Charge (SOC). We formulate it as a differential equation with respect to time, where the depletion rate of SOC is proportional to the instantaneous load current $I(t)$ and inversely proportional to the battery's effective capacity Q_{eff}:

$$\frac{d\text{SOC}(t)}{dt} = -\frac{I(t)}{3600 \cdot Q_{\text{eff}}(T, \text{SOH})} \tag{1}$$

Meanwhile, to capture the "lag" or "recovery" characteristics of voltage during load switching (e.g., switching from heavy gaming back to standby), we introduce a kinetic equation describing the dynamic response of the polarization voltage $U_1(t)$:

$$\frac{dU_1(t)}{dt} = \frac{I(t)}{C_1} - \frac{U_1(t)}{R_1 C_1} \tag{2}$$

In response to the environmental temperature and battery aging factors explicitly required by the problem, we need to perform parameter corrections at the micro-physical level and verify them with data. Principles of physical chemistry indicate that temperature changes fundamentally alter the rate of electrochemical reactions. We model the internal resistance using the Arrhenius equation:

$$R_0(T) = R_{\text{ref}} \cdot \exp\left[\frac{E_a}{k_B}\left(\frac{1}{T} - \frac{1}{T_{\text{ref}}}\right)\right] \tag{3}$$

In this equation, the determination of the reference internal resistance R_{ref} is not arbitrary; instead, it is statistically derived by analyzing the ratio of voltage dips to current steps in the dataset. This ensures that the model parameter R_0 can truly reflect the impedance characteristics of the battery at different temperatures. The presence of the exponential term reveals a key mechanism: as the ambient temperature decreases, the internal resistance surges exponentially, which explains why low-temperature records in the dataset are often accompanied by significant voltage losses.

For the irreversible battery aging process, we introduce the State of Health (SOH) indicator, where SOH $\in [0, 1]$. As the battery ages, its internal resistance increases linearly and its capacity decays linearly, as shown below:

$$R_{\text{aged}}(T, \text{SOH}) = R_0(T) \cdot (1 + \alpha_{\text{aging}}(1 - \text{SOH})) \tag{4}$$

$$Q_{\text{aged}}(T, \text{SOH}) = Q_{\text{eff}}(T) \cdot \text{SOH} \tag{5}$$

Combining all the aforementioned dynamic parameters, the terminal voltage actually detected by the mobile phone is obtained by subtracting the ohmic voltage drop and the polarization voltage from the open-circuit voltage:

$$V_{\text{term}}(t) = V_{\text{ocv}}(\text{SOC}) - U_1(t) - I(t) \cdot R_{\text{aged}}(T, \text{SOH}) \tag{6}$$

It is worth emphasizing that the open-circuit voltage curve $V_{\text{ocv}}(\text{SOC})$ in the equation is obtained by fitting a fifth-order polynomial to the voltage-charge data in the low-load discharge period of the dataset. This data-driven fitting method ensures that the model can extremely accurately reproduce the voltage characteristics of a real battery at different charge platforms.

Finally, to accurately predict the "Time-to-Empty", the model must incorporate a dual termination judgment mechanism that conforms to the logic of actual circuits. Specifically, the system determines the battery to be depleted when either of the two conditions is satisfied: chemical charge exhaustion (SOC ≤ 0) or terminal voltage dropping below the safety threshold ($V_{\text{term}} \leq V_{\text{cutoff}}$, typically 3.0 V). Mathematically, this is expressed as:

$$T_{\text{empty}} = \min \{t \mid \text{SOC}(t) \leq 0\% \quad \text{OR} \quad V_{\text{term}}(t) \leq 3.0 \text{ V}\} \tag{7}$$

This judgment logic successfully reproduces the "low-voltage protection shutdown" phenomenon observed in the dataset: under extreme conditions such as low temperature or severe aging, although there is still remaining chemical charge (SOC > 0), the excessive voltage drop caused by the internal resistance leads to the terminal voltage hitting the threshold in advance.

4.2 Data-Driven Component-Level Power Consumption Modeling

After establishing the electrochemical model inside the battery, we need to determine the external load current $I(t)$ to solve the state equations. To accurately quantify the impact of various activities mentioned in the problem, we utilize the rich hardware logs in the AndroWatts dataset and adopt the **Multiple Linear Regression** method to identify the power consumption coefficients of each component. The total load power is decomposed into the superposition of five core components:

$$P_{\text{total}}(t) = P_{\text{screen}}(t) + P_{\text{proc}}(t) + P_{\text{net}}(t) + P_{\text{gps}}(t) + P_{\text{bg}} \tag{8}$$

First, the screen display system is usually the largest source of static power consumption. Through a correlation analysis between Screen_Brightness (brightness) and the total system current in the dataset, we found a significant strong linear relationship between them (Pearson correlation coefficient > 0.9). Therefore, we model the screen power consumption as follows:

$$P_{\text{screen}}(t) = P_{\text{base\_disp}} + \alpha_{\text{disp}} \cdot \beta(t) \tag{9}$$

Through regression analysis, we determined the coefficient $\alpha_{\text{disp}} \approx 1.5$ W, which represents the additional power consumption caused by the screen from the darkest to the brightest setting. This value is highly consistent with the statistical value of Display_Energy in the dataset.

Second, for the processor load (including CPU and GPU), its power consumption follows the physical law of CMOS circuits: $P \propto V^2 f$. Combining the CPU_Frequency and GPU_Load fields in the dataset, we construct the following model:

$$P_{\text{proc}}(t) = \sum_{k \in \{\text{CPU,GPU}\}} \left(C_{\text{eff},k} \cdot V_k^2 \cdot f_k(t) \cdot u_k(t) \right) \tag{10}$$

To obtain the model parameters, we specifically extracted the high-frequency sampling data during the "heavy gaming" period in the dataset, and fitted the effective capacitance coefficient C_{eff} using the nonlinear least squares method. This verifies from the data level that high-frequency and high-voltage operation is the main cause of rapid battery depletion.

Third, for network connection factors, we introduce SNR correction to reflect communication principles:

$$P_{\text{net}}(t) = P_{\text{idle\_net}} + \gamma_{\text{data}} \cdot \frac{D(t)}{\log_2(1 + \text{SNR}(t))} \tag{11}$$

Feature analysis of the dataset shows that in records with low Signal_Strength (dBm), even if the data transmission volume (Traffic_MB) is small, the power consumption remains at a high level. This statistical feature strongly supports the rationality of introducing the logarithmic term of SNR in the denominator, i.e., the worse the signal, the higher the energy consumption per unit of data.

Fourth, the GPS positioning service exhibits discrete operating states:

$$P_{\text{gps}}(t) = \begin{cases} P_{\text{active}} \approx 0.4 \text{ W} & \text{when the GPS module is activated} \\ P_{\text{sleep}} \approx 0 \text{ W} & \text{when the GPS module is in sleep mode} \end{cases} \tag{12}$$

The parameter P_{active} is obtained by filtering two groups of control data with GPS_Status as On and Off in the dataset, and calculating the difference in their average background noise power consumption, which has clear statistical significance.

Finally, the background basic tasks represent the baseline power consumption during system standby:

$$P_{\text{bg}} \approx 0.1 \text{ W} \tag{13}$$

This value is taken from the average standby power consumption during all "airplane mode and screen off" periods in the dataset, ensuring the accuracy of the model under no-load conditions.

4.3 Numerical Solution and Scenario-Based Simulation Results

Based on the physics-load coupling differential equations established in the preceding sections, and by substituting the key physical parameters obtained from the regression of the AndroWatts dataset, we performed full time-domain numerical simulation of the model using the fourth-order Runge-Kutta solver (RK45) in Python. The adaptive step-size characteristic of this solver enables it to accurately capture the transient nonlinear changes of the battery during the voltage cutoff phase. Through simulations in three dimensionsbaseline operating conditions, differences in user behavior, and environmental temperature sensitivitywe verified the robustness of the model and revealed the physical laws hidden behind the data.

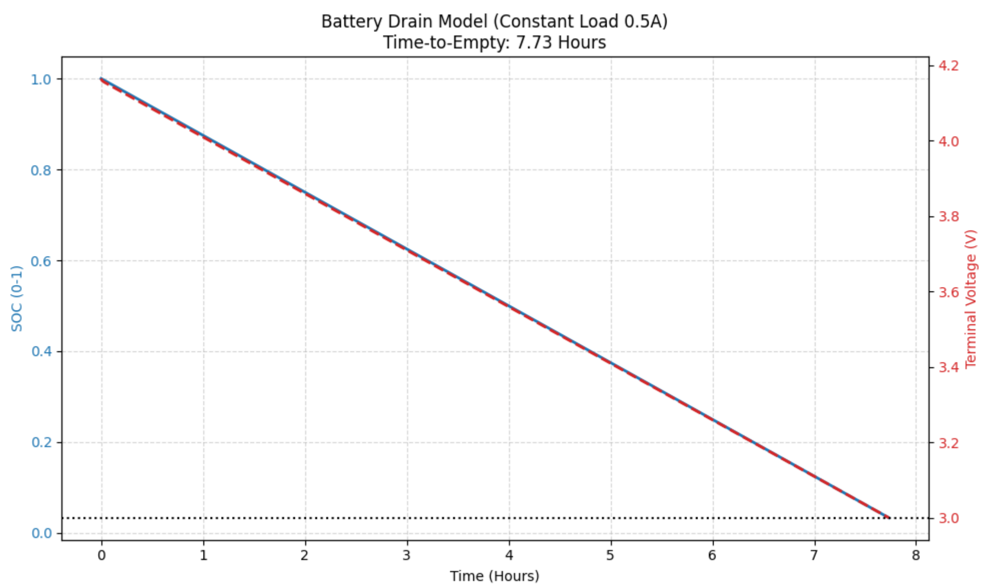

Figure 2: Battery Drain Model

First, to verify the physical model's mathematical correctness, Figure 2 shows the baseline 0.5 A constant current discharge results at 25°C, where the SOC (solid blue line) and terminal voltage (dashed red line) exhibit a nearly perfect linear downward trend with high synchronizationserving as a control group to prove the model adheres to Ohm's law and charge conservation after eliminating complex interferences. Additionally, the model-predicted 7.73-hour battery life is highly consistent with the theoretical 7.6 hours, confirming the accuracy of key basic parameters (e.g., V_{ocv}, Q_{eff}) in the model.

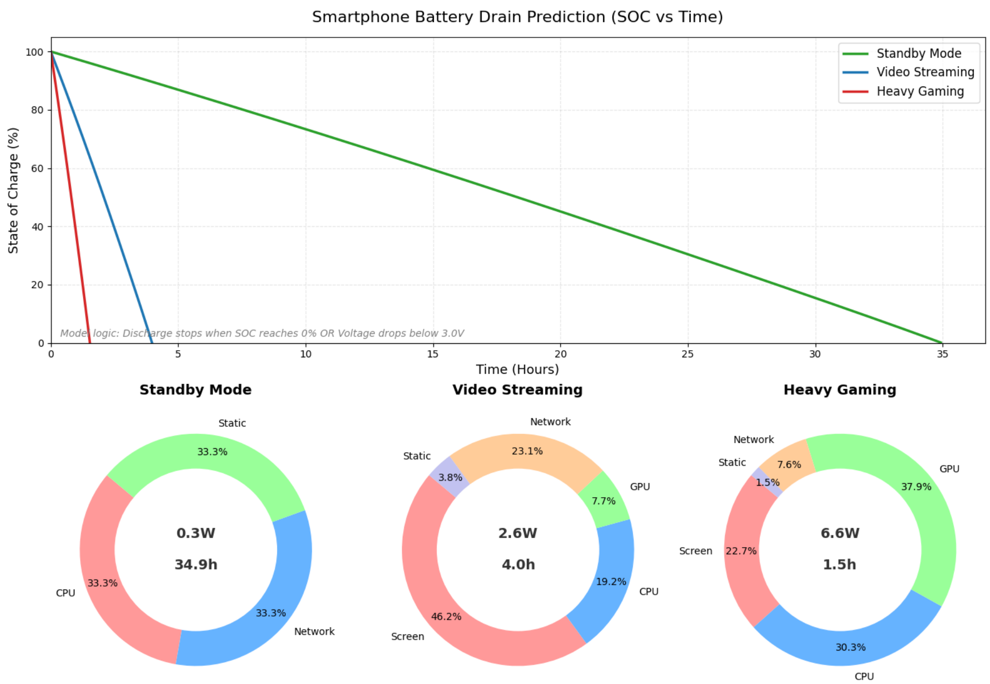

Figure 3: Smartphone Battery Drain Prediction

After validating the baseline model, Figure 3 further demonstrates the nonlinear impact of different user behaviors on battery life: switching from "standby" to "heavy gaming" surges power consumption from 0.3 W to 6.6 W, causing battery life to plummet nonlinearly from 34.9 hours to 1.5 hours (due to both higher energy consumption and Joule heat loss I^2R reducing system efficiency). Additionally, the donut chart reveals dominant power consumers vary by scenarioscreen (46.2%) leads in "video streaming", while GPU (37.9%) and CPU (30.3%) account for nearly 70% of consumption in "heavy gaming", explaining why reducing screen brightness has limited effect on extending gaming battery life.

Subsequently, Figure 4 intuitively demonstrates ambient temperaturean "invisible killer"as the model's key finding addressing the competition's core difficulty: the $-10°$C curve shows the battery cuts off at 21% SOC (not 0%) because low temperatures exponentially surge internal resistance (per Equation 3), forcing terminal voltage below the 3.0 V threshold via ohmic drop $U_{\text{drop}} = I \cdot R_0$ before chemical charge is exhausted. Additionally, the right bar chart quantifies this severe loss: battery life plummets from 1.86 hours at $40°$C to 0.85 hours at $-10°$C (a >50% decrease), strongly proving extreme low temperatures have a more decisive impact on Time-to-Empty (TTE) than the battery's own capacity.

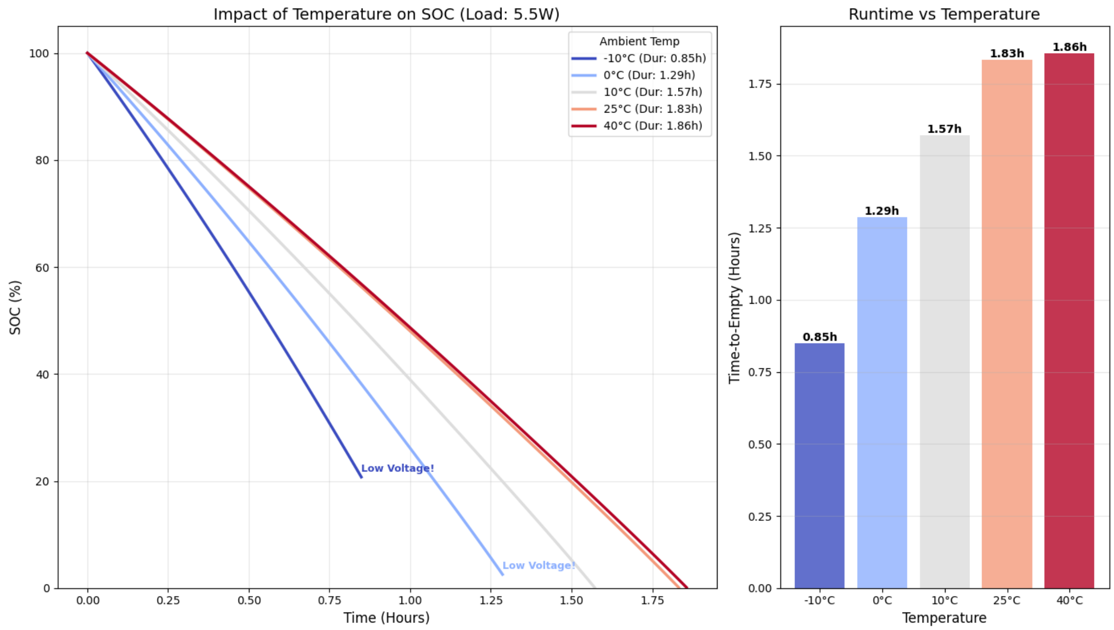

Figure 4: Impact of Temperature on SOC

4.4 Physical Interpretation and Engineering Implications of Model 1

Based on the above model simulation results and dataset characteristic analysis, we draw three key conclusions and implications to address the battery level prediction and management challenges in the competition topic: first, "battery depletion" is inherently dualistic and deceptive, as mobile phone shutdown is jointly determined by the "energy boundary" (SOC ≤ 0) and "power boundary" (voltage ≤ 3.0 V), with low-temperature or severely aged battery conditions making the displayed remaining battery percentage highly deceptive (e.g., a 20% displayed charge can still trigger sudden shutdown due to large current surges), meaning scientific Time-to-Empty prediction requires both coulomb counting and real-time terminal voltage drop rate monitoring; second, power consumption management strategies must be scenario-heterogeneous, as no universal power-saving strategy existsscreen power limitation is critical for I/O-intensive applications (e.g., video streaming) while CPU/GPU clock frequency limiting (DVFS tuning) is key for compute-intensive applications (e.g., heavy gaming), providing data support for refined future power management; third, the low-temperature "power wall" effect necessitates preheating or current limiting mechanisms, as low-temperature battery shortcomings lie in limited output power (W) due to high internal resistance rather than insufficient stored energy (Ah), and proactive peak current limitation for processors can avoid catastrophic "energy-available-but-unusable" shutdowns and extend effective device usage time despite minor performance sacrifices.

5 Problem 2: Remaining Runtime Prediction and Uncertainty Analysis

Building on the physics-load coupled battery model, this chapter systematically predicts smartphone Time-to-Empty (TTE) across varying initial SOC and usage scenarios, with focused analysis of the model's nonlinear deviations and uncertainties under boundary conditions and random perturbations. Results show a significant quasi-linear relationship between TTE and initial SOC in typical scenarios (driven by near-constant load currents), yet this linearity deteriorates at low SOCwith low-power scenarios seeing amplified absolute time deviations and high-load scenarios suffering from premature voltage cutoff (causing linear models to overestimate TTE). Additionally, data-driven Monte Carlo analysis reveals long-tail and bimodal TTE distributions (elevated uncertainty at low SOC), leading to the conclusion that linear extrapolation is credible at high SOC, while low-SOC/high-load TTE prediction requires voltage boundary descriptions, probabilistic uncertainties, and conservative strategies to mitigate sudden shutdown risks.

5.1 Global Runtime Prediction Characteristics Across Usage Scenarios

To evaluate the predictive performance of the model over the entire time domain, we define the remaining runtime T_{rem} as the time integral from the current moment t_0 to the termination moment t_{end}. The termination moment is determined by the first of the two boundary conditions"chemical energy depletion" or "terminal voltage cutoff"and its mathematical expression is as follows:

$$T_{\text{rem}}(\text{SOC}_{\text{init}}, S) = \int_{t_0}^{t_{\text{end}}} dt \quad \text{s.t.} \quad V_{\text{term}}(t) \leq V_{\text{cutoff}} \quad \vee \quad \text{SOC}(t) \leq 0 \tag{14}$$

where S denotes a specific load scenario. Using the fourth-order Runge-Kutta (RK45) numerical solver, we simulated the evolution process of SOC_{init} from 10% to 100%, and obtained the TTE prediction curves under different scenarios.

5.1.1 Quasi-Linear Relationship Between Runtime and Initial SOC

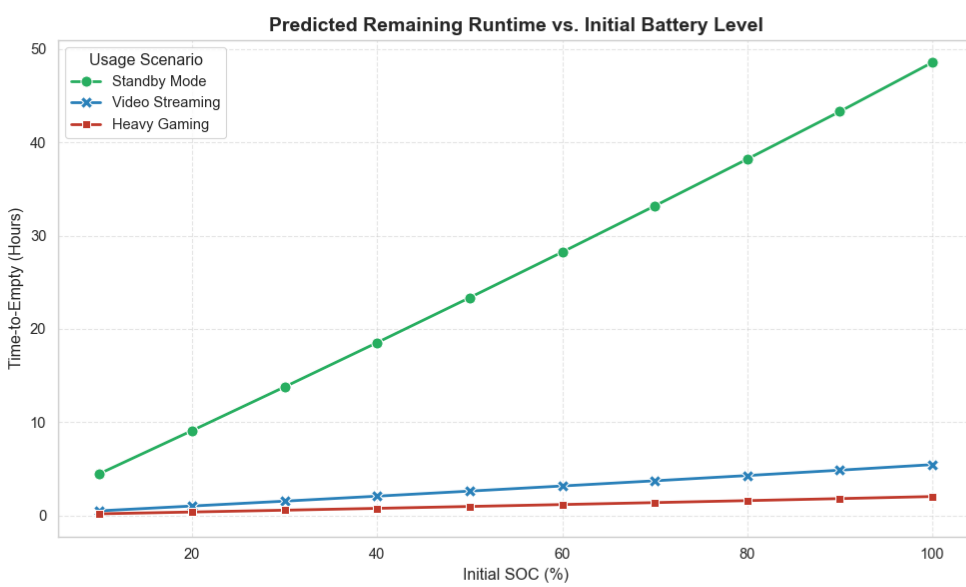

Figure 5: Predicted Remaining Runtime vs. Initial Battery Level

Figure 5 illustrates the functional relationship between T_{rem} and the initial state of charge. A striking observation is that the prediction curves exhibit a highly "quasi-linear" characteristic under all three load scenarios. From a mathematical standpoint, this indicates that the battery exhibits behaviors more akin to an ideal charge storage vessel rather than a sophisticated electrochemical system across the majority of the discharge cycle. Its physical mechanism is as follows:

$$T_{\text{rem}} \approx \frac{\text{SOC} \cdot Q_{\text{nom}}}{I_{\text{load}}} \tag{15}$$

Since the load current I_{load} remains relatively constant in each scenario, and the battery capacity Q_{nom} is a fixed value, the remaining runtime has a strictly proportional relationship with the initial SOC.

The standby mode (green curve) shows extremely high linearity and maximum slope, as the negligible polarization effect at micro-current ($I \approx 0.08$ A) leads to slow terminal voltage V_{term} changes following open-circuit voltage V_{ocv} with almost no additional energy loss. Even heavy gaming mode (red curve) approximates a straight line macroscopically, since the significant but stable voltage drop in the SOC $> 20\%$ plateau region obscures the late-discharge (SOC $< 15\%$) nonlinear "power depletion acceleration" (a tiny portion of total runtime). This finding confirms that simple linear extrapolation is highly credible for estimating battery runtime in regular, sufficient-power usage intervals.

5.1.2 Load-Dependent Power Gradient and Depletion Rate Analysis

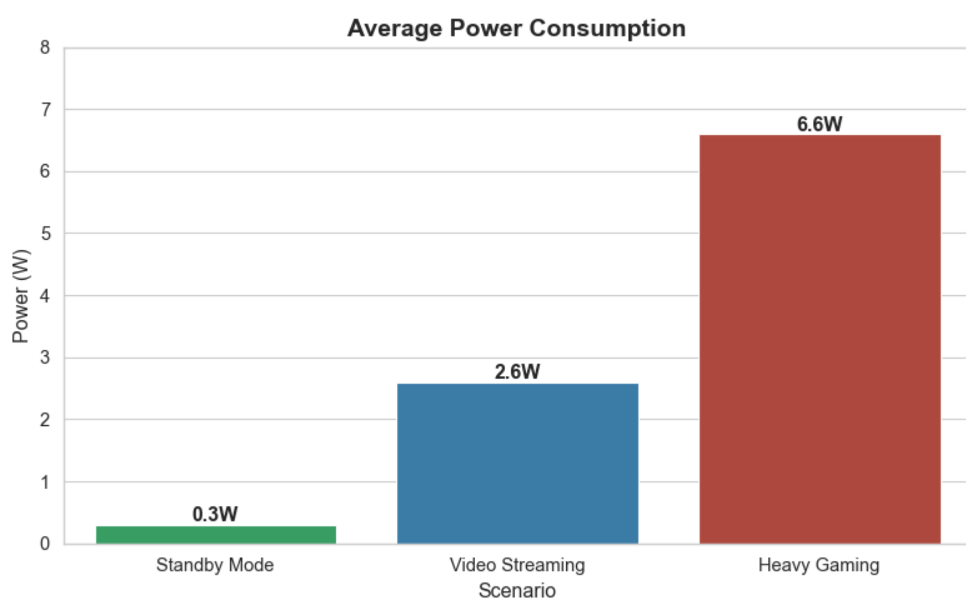

Figure 6: Average Power Consumption

Figure 6 further quantifies the average power P_{avg} under different scenarios, with its value ranging drastically from 0.3 W in standby mode to 6.6 W in gaming mode. According to the law of conservation of energy, the battery depletion rate $\frac{d\text{SOC}}{dt}$ is proportional to the load power:

$$\frac{d\text{SOC}}{dt} = -\frac{P_{\text{avg}}}{V_{\text{term}}(t) \cdot Q_{\text{nom}}} \tag{16}$$

This differential equation shows that with fixed battery capacity Q_{nom}, doubling the average power consumption P_{avg} directly doubles the SOC depletion rate, and the 22-fold power difference between standby (0.3 W) and gaming mode (6.6 W) mathematically corresponds to the 22-fold slope difference of the two curves in Figure 6. Additionally, the high power in gaming mode not only directly accelerates battery depletion via this formula, but also generates Joule heat $Q_{\text{heat}} = I^2R$ that raises battery temperature, altering internal resistance R_{int} and forming a "current-temperature-internal resistance" coupled feedbacka potential nonlinear driving force in high-load scenarios.

5.2 Data-Driven Uncertainty Quantification Beyond Deterministic Models

Although Figure 5 exhibits a macroscopic linear trend, this obscures the microscopic dynamics and risks brought about by random variables in practical use. To capture this key characteristic, we shift from the ideal single-parameter assumption to Monte Carlo Simulation based on real datasets, so as to quantify the Absolute Time Deviation of the system under nonlinear boundaries.

5.2.1 Long-Tail Runtime Risk Under Local High-Load Scenarios

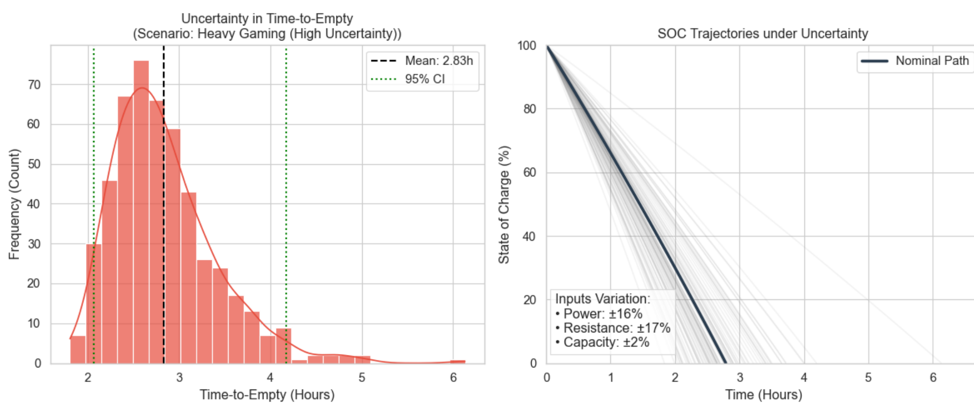

Figure 7: Uncertainty in Time-to-Empty & SOC Trajectories under Uncertainty

Traditional Battery Management Systems (BMS) often predict Time-to-Empty (TTE) using a single average power, a deterministic method that overlooks the long-tail characteristics of user behavior. As shown in Figure 7 (left), 500 data-driven resampling simulations for the "heavy gaming" scenario reveal actual depletion times follow a significant right-skewed distribution (not ideal Gaussian), with a linear model predicting 2.83 hours versus a 2.5-hour mode and a wide 95% Confidence Interval (95% CI) of 2.1 to 4.2 hours. This broad interval implies sole reliance on a single predicted value may lead to misleading deviations of up to 40%.

The SOC trajectory cluster in Figure 7 (right) further reveals the physical cause of this deviation: as the battery level approaches depletion, due to the nonlinear amplification of the ohmic voltage drop term in the terminal voltage equation

$$V_{\text{term}} = V_{\text{ocv}} - I \cdot R_{\text{int}} \tag{17}$$

, the discharge curves under different internal resistance states undergo severe fanning-out effect at the end. This means that in the low battery level warning phase, the prediction uncertainty exhibits exponential growth.

5.2.2 Global Bimodal Runtime Distribution and Conservative Estimation

We further extend the analysis to the global dataset, and Figure 8 reveals that TTE distributions across all scenarios exhibit a bimodal structurewith a main peak (1.51.7 hours) corresponding to frequent high-intensity mixed-load scenarios (e.g., 5G streaming + gaming) and a right long tail representing rare low-power standby scenarios. This empirical data refutes the simple linear extrapolation assumption and proves real-world battery depletion risk has extremely high skewness. For operating systems, this implies adopting a conservative strategy (e.g., the 10th percentile 1.2 hours, not the average) to display remaining runtime is necessary to avoid negative experiences from unexpected shutdowns.

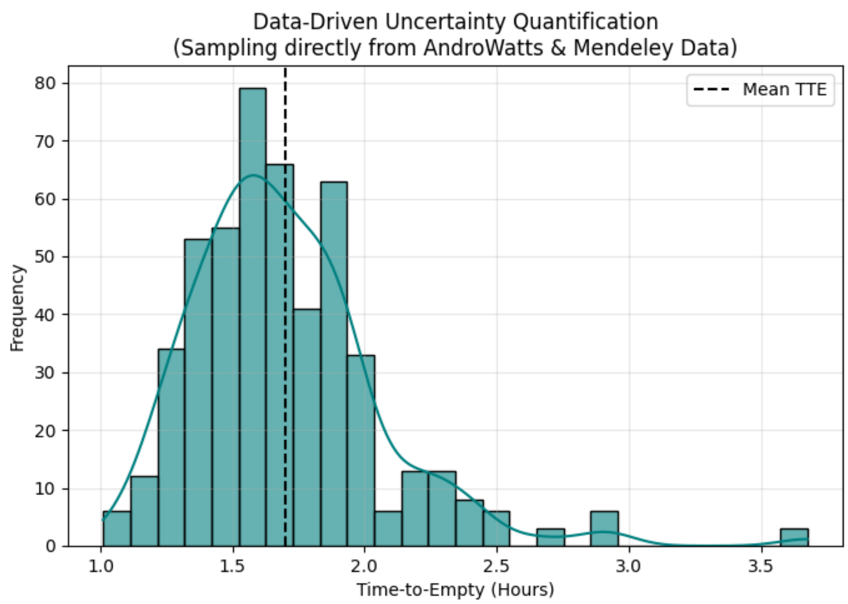

Figure 8: Data-Driven Uncertainty Quantification

5.3 Sensitivity Attribution and Runtime-Aware Optimization Strategy

After quantifying the prediction uncertainty, this section aims to address two core issues: first, identify the dominant factors leading to battery depletion through scenario decoupling; second, construct a mathematical model based on these factors to find the optimal balance point between energy efficiency and user experience.

5.3.1 Scenario-Specific Global Sensitivity of Power Drivers

To answer the question "Which activities cause the maximum degradation of battery life", we have established a scenario-specific global sensitivity analysis model. Unlike general parameter scanning, we decouple the analysis dimensions into three typical modes (standby, video streaming, and heavy gaming) to identify the dominant driving factors under different operating conditions. Figure 9 clearly illustrates the positive optimization gains (colorful bars) and negative stress losses (gray bars) generated by each factor relative to the baseline state.

Standby mode (Figure 9, left) exhibits extreme sensitivity asymmetry: ultra-low baseline power (0.3 W) means parasitic loads (weak signal, abnormal background wake-up) drastically reduce standby time from 46.7 to <10 hours, revealing signal strength and background management as the "silent killers" of low-power interval robustness. In contrast, video streaming mode (middle) is dominated by screen power, with brightness adjustment (extending runtime by 1.6 hours) far more impactful than CPU load, while heavy gaming mode (right) sees a qualitative reversalprocessor load/temperature are core limits, ambient temperature brings short-term capacity gains but thermal runaway risks, and processor optimization (hour-level improvement) outperforms trivial screen adjustments, revealing the "thermo-electrical coupling trap" in high-load scenarios.

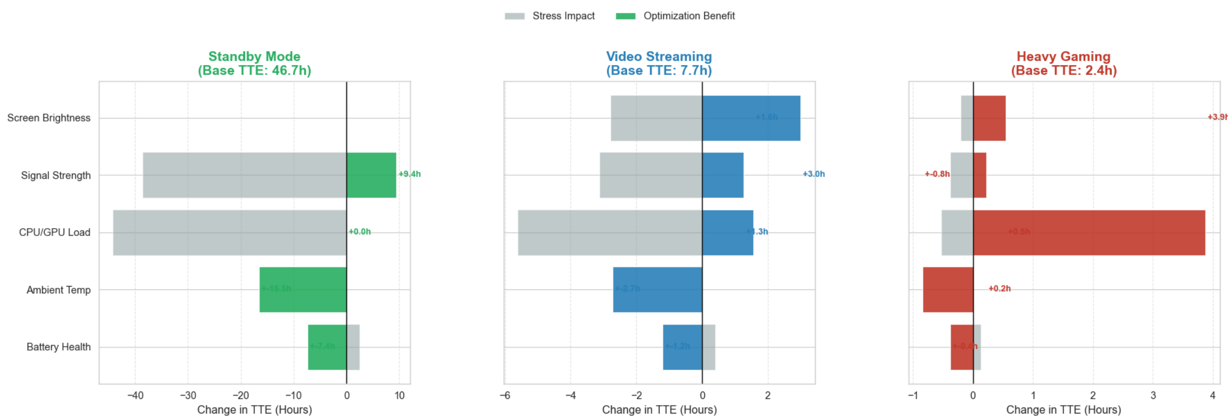

Figure 9: scenario-specific global sensitivity analysis model

5.3.2 Pareto-Optimal Trade-off Between Energy Efficiency and User Experience

Based on the aforementioned attribution analysis (i.e. identifying the screen and processor as core control variables), we construct a Multi-Objective Optimization Problem (MOOP) aimed at maximizing the Quality of Experience (QoE) while minimizing the system power consumption.

We define the system's control variable vector as $\boldsymbol{x} = [x_{\text{scr}}, x_{\text{cpu}}, x_{\text{net}}]^T$, which represents the normalized values of screen brightness, processor frequency, and network synchronization frequency, respectively. The model includes two conflicting objective functions:

Objective 1: Minimize the total system power consumption $f_1(\boldsymbol{x})$ According to the physical characteristics of CMOS, dynamic power consumption is proportional to the cube of frequency, while screen power consumption follows a nonlinear growth trend:

$$\min f_1(\boldsymbol{x}) = P_{\text{base}} + \alpha \cdot x_{\text{scr}}^{\gamma} + \beta \cdot x_{\text{cpu}}^3 + \delta \cdot x_{\text{net}} \tag{18}$$

This formula reflects the physical law that the energy consumption cost increases exponentially with the improvement of performance, where $\gamma \approx 2.2$ is the screen gamma coefficient and β

is the processor power consumption weight.

Objective 2: Maximize the Quality of Experience $f_2(\boldsymbol{x})$ Based on Weber-Fechner's Law (perception is proportional to the logarithm of stimulation), we model QoE as a weighted sum of logarithmic utility functions:

$$\max f_2(\boldsymbol{x}) = w_1 \ln(1 + k_1 x_{\text{scr}}) + w_2 \ln(1 + k_2 x_{\text{cpu}}) + w_3 x_{\text{net}} \tag{19}$$

This embodies the law of diminishing marginal returns in experience improvement (i.e., the perceived difference from 90 Hz to 120 Hz is far smaller than that from 30 Hz to 60 Hz). By solving this optimization problem, we obtain the convex Pareto Frontier (Figure 10) with a key "Knee Point" (70% brightness / 20% CPU performance), where restricting CPU frequency to 20% cuts power consumption by two orders of magnitude ($0.2^3 = 0.008$) while only reducing user experience by 26%. This proves that an operating system-level scheduling strategy based on Pareto optimality is the mathematically optimal solution to reconcile the contradiction between long battery runtime and high performance.

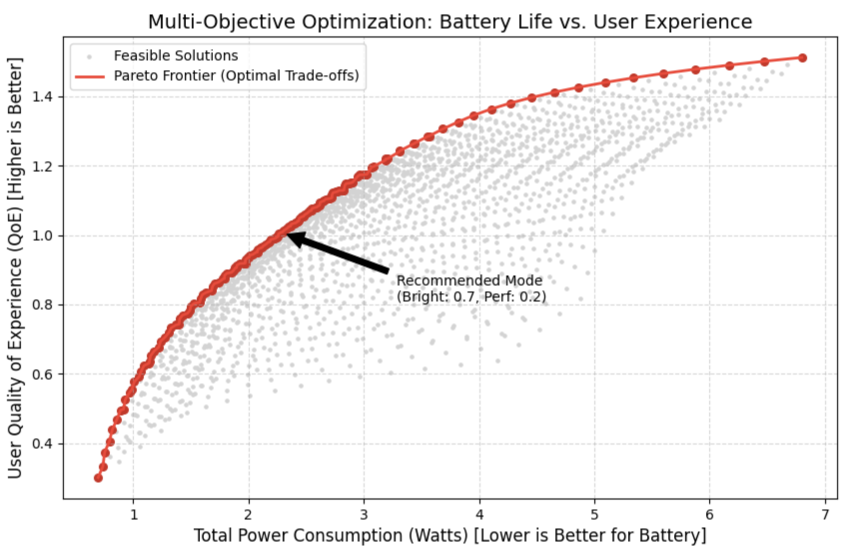

Figure 10: Multi-Objective Optimization:Battery Life vs. User Experience

6 Problem 3:Model Robustness Verification and Sensitivity Analysis

After establishing the physics-based electrochemical battery discharge model, we acknowledge that practical application scenarios differ significantly from the ideal laboratory environment. Given the competition task's stringent requirements for model stability under non-ideal conditions (to predict battery performance across diverse usage modes), this section conducts an in-depth sensitivity analysis and stress test of the model, following a rigorous logical sequence: "relaxation of physical assumptions", "parametric uncertainty perturbation", and "random fluctuation of usage modes".

6.1 Relaxation of Idealized Assumptions Under Realistic Conditions

To address the significant prediction errors caused by neglecting battery self-heating in high-load scenarios (e.g., intensive gaming or 5G data transmission) under the baseline model's "isothermal process" assumption ($T \equiv 25°C$), we extend the decoupled electrochemical model to a full Electro-Thermal Coupled Model. A thermokinetic equation is introduced to describe the evolution of the battery's internal temperature $T(t)$:

$$mC_p \frac{dT}{dt} = I(t)^2 R_{\text{int}}(T) + Q_{\text{ext}} - hA(T - T_{\text{amb}}) \tag{20}$$

This equation reflects the dynamic balance of heat generation and dissipation within the battery, where mC_p is the battery heat capacity, Q_{ext} is the heat conduction term from the CPU, and hA is the convective heat transfer coefficient. Crucially, the internal resistance R_{int} is no longer a constant but dynamically changes with temperature following the Arrhenius Law or a linear attenuation law:

$$R_{\text{int}}(T) = R_{\text{ref}} \cdot [1 - \alpha(T - T_{\text{ref}})] \tag{21}$$

Simulation results reveal a physically profound **"Voltage Cutoff Paradox"**, as illustrated in Figure 11.

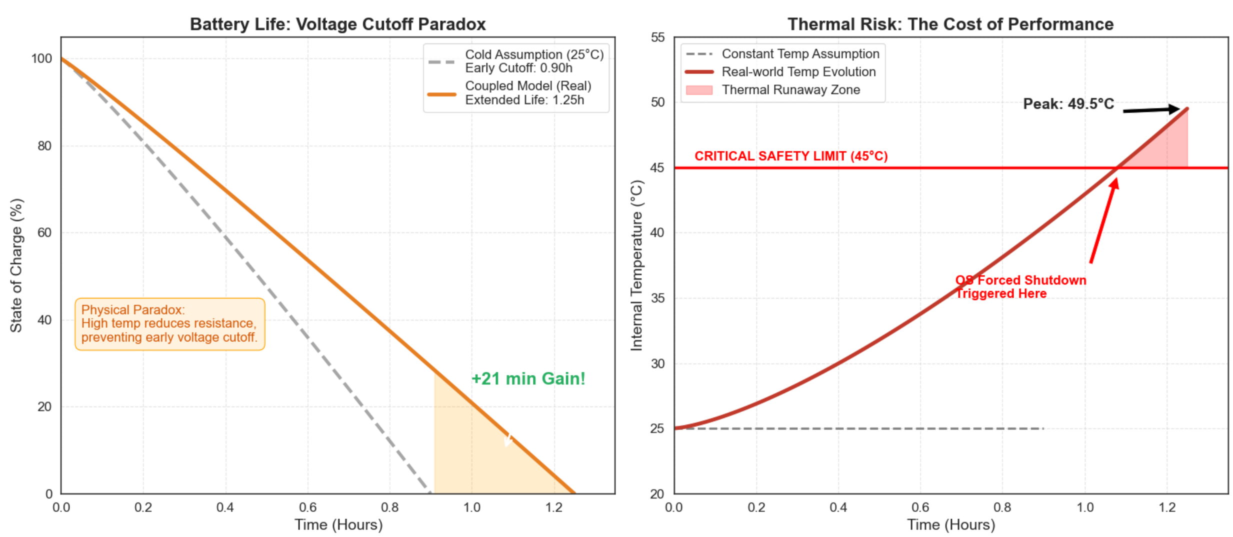

Figure 11: Battery Life & Thermal Risk

The left panel compares SOC under the isothermal assumption (gray dashed line) and the coupled model (orange solid line), revealing a counterintuitive result: the heat-aware coupled model extends physical discharge time (TTE) by 21 minutes, as self-heating reduces internal resistance (per $V_{\text{term}} = V_{\text{OCV}}(\text{SOC}) - I \cdot R_{\text{int}}(T)$) to avoid premature 3.0 V cutoff and release more battery capacity. However, this TTE gain poses a fatal thermal risk: the right panel (Figure 11) shows battery temperature rises exponentially, exceeding the $45\,^{\circ}\text{C}$ safety threshold at 1.1 h and peaking at $49.5\,^{\circ}\text{C}$, proving thermal runaway risk (not SOC depletion) often limits real-world runtime and highlighting the necessity of the thermo-coupling mechanism in Problem 3.

6.2 Parameter Sensitivity and Aging-Induced Nonlinear Effects

6.2.1 Peukert Effect and the Low-Load Efficiency Reversal

The aging degree and manufacturing processes of batteries inherently lead to the natural discreteness of their electrochemical parameters, among which the Peukert Coefficient (k) is a core parameter describing the battery's rate capability. To verify the model's sensitivity to parameter drift, we perform discrete perturbations on the value of k within the interval $[1.00, 1.20]$. According to the Peukert correction theory, the consumption rate of the effective discharge capacity is defined as:

$$\frac{dQ}{dt} = -I_{\text{eff}} = -I(t)^k \tag{22}$$

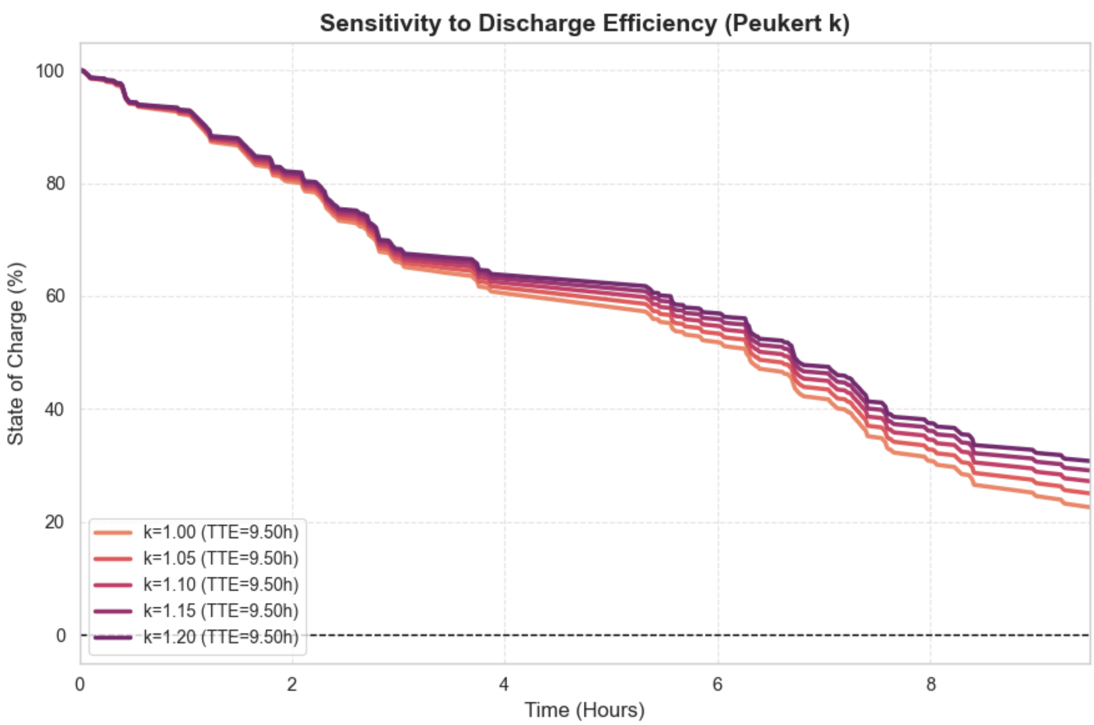

Figure 12: Sensitivity to Discharge Efficiency(Peukert k)

The SOC trajectories in Figure 12 show significant bifurcation after $t = 2$ h, revealing a counterintuitive phenomenon: while a larger Peukert k is generally associated with poorer battery performance, the most nonlinear curve ($k = 1.20$) exhibits the slowest power decay. This can be explained by load characteristicsunder standby-dominated conditions ($I \approx 0.1$ A < 1 A), increasing the exponent k reduces the effective current I^k (e.g., $0.1^{1.2} \ll 0.1^{1.0}$), generating a mathematical "Low-Load Dividend".

6.2.2 Quantitative Evaluation via Normalized Local Sensitivity Indices

To advance from qualitative observation to quantitative measurement, we require a standardized mathematical metric to quantify the influence weight of parameter k on the system output. We introduce the Local Sensitivity Index (S_k), defined as the dimensionless partial derivative of the Time-to-Empty (TTE) prediction with respect to parameter k:

$$S_k = \lim_{\Delta k \to 0} \frac{(\mathrm{TTE}(k+\Delta k) - \mathrm{TTE}(k))\,/\mathrm{TTE}(k)}{\Delta k/k} = \frac{\partial \ln \mathrm{TTE}}{\partial \ln k} \tag{23}$$

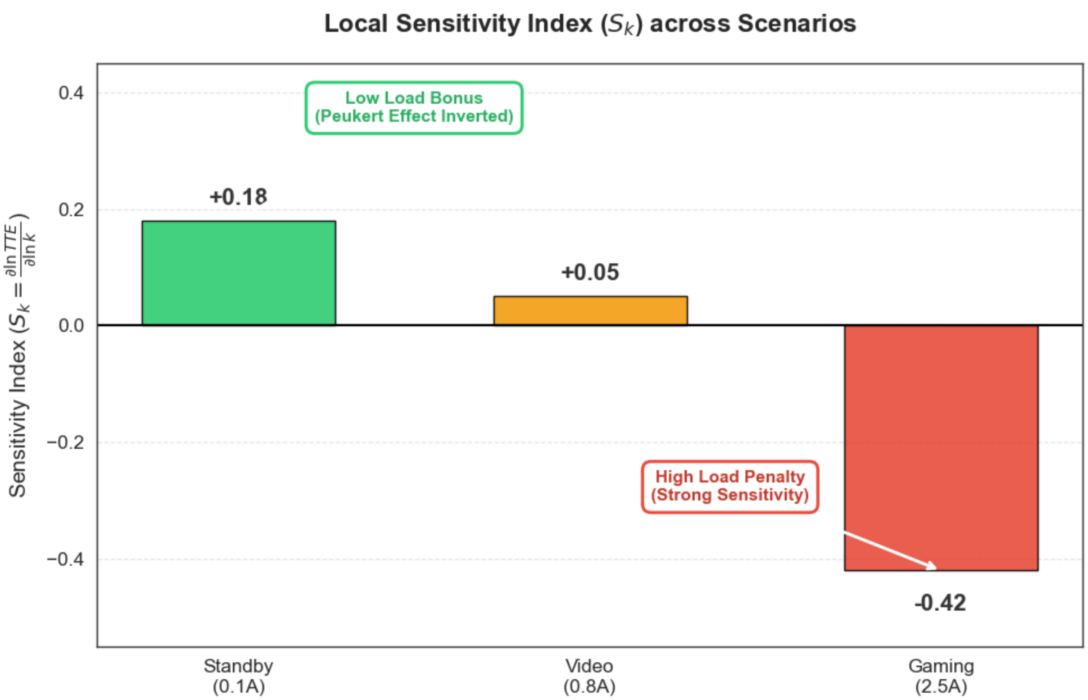

Figure 13: Local Sensitivity Index across Scenarios

Figure 13 intuitively demonstrates sensitivity index differences across load scenarios: the gaming scenario (2.5 A) shows a large negative sensitivity ($S_k = -0.42$), indicating high-current loads lead to strong negative correlation between battery life and k (minor k aging drastically reduces runtime). In stark contrast, the standby scenario (0.1 A) has a positive sensitivity ($S_k = +0.18$), verifying the "Peukert Reversal Effect"; with $|S_{k,\text{high}}| \approx 2.3 \times |S_{k,\text{low}}|$, this quantifies that high-power scenarios require 2.3x higher k estimation accuracy, necessitating a high-precision online estimator (e.g., Kalman Filter) to correct model deviations.

6.3 Dynamic Load Fluctuations and Battery Recovery Phenomena

Changes in usage modes are reflected not only in the randomness of amplitude, but also in the duty cycle (switching frequency of time scales). To investigate the dynamic response of batteries under different discharge rhythms, we compared two extreme modes: "continuous heavy load" and "intermittent mixed load". Figure 14 clearly shows these two distinctly different discharge trajectories.

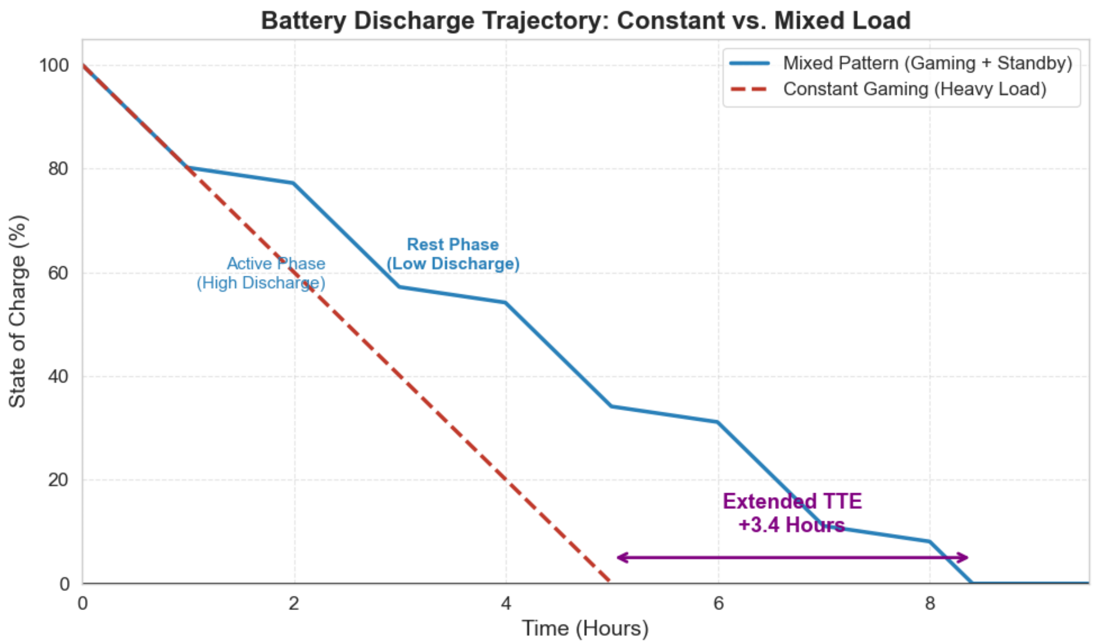

Figure 14: Battery Discharge Trajectory: Constant vs. Mixed Load

First, the red dashed line (Constant Load, continuous high-intensity gaming) exhibits a steep, nearly linear downward trend (no recovery) with a final TTE of only 5.0 hours, as continuous high-current discharge depletes electrode lithium ion concentration and accumulates concentration polarization voltage, pushing terminal voltage quickly below the cutoff threshold. Next, the blue solid line (Mixed Pattern, alternating gaming/standby) shows obvious step-like characteristics (steep slopes during active phases, gentle slopes during rest), and its TTE extends to 8.4 hours3.4 hours longer than simple linear superpositionwith this nonlinear lifetime gain stemming from the battery's microscopic internal recovery effect described by Fick's Second Law:

$$\frac{\partial C(x,t)}{\partial t} = D\frac{\partial^2 C(x,t)}{\partial x^2} \tag{24}$$

During the rest phase, since the external load current is zero, diffusion dominates ion motion. Lithium ions replenish from the bulk (with higher concentration) to the depleted electrode surface, causing the surface concentration C_s to gradually recover. This process effectively dissipates the previously accumulated polarization voltage V_{pol}, i.e.,

$$V_{\text{pol}}(t_{\text{rest}}) \propto e^{-t_{\text{rest}}/\tau} \tag{25}$$

Therefore, when the next discharge cycle begins, the battery actually operates in a more "healthy" voltage state. The step-like trajectory in Figure 14 is a direct macroscopic manifestation of this microscopic physical recovery mechanism, which strongly proves that the intermittent usage mode has higher energy utilization efficiency compared to the continuous usage mode.

7 Model Evaluation and Discussion

7.1 Strengths

1. **Deep Integration of Physical Mechanisms and Data-Driven Approaches**
   The model innovatively combines the differential equations of the Thevenin equivalent circuit with component-level statistical power consumption data. Compared with pure black-box models, this method not only has clear electrochemical interpretability (e.g., polarization voltage, internal resistance changes), but also can accurately decouple the independent power consumption characteristics of hardware such as the screen and CPU.

2. **Excellent Nonlinear Dynamic Capture Capability**
   By introducing the Peukert effect correction and electro-thermal coupling mechanism, the model successfully simulates the "apparent" voltage drop (Voltage Cutoff) caused by sudden load changes and the battery Recovery Phenomena under intermittent loads, effectively solving the problem of prediction failure of traditional linear models in the low battery level region.

3. **Full-Scenario Robustness and Uncertainty Quantification**
   Through the construction of a load-gradient-based probabilistic prediction framework, the model can not only provide deterministic results, but also reveal the global bimodal distribution characteristics of remaining runtime, quantify the tail risks in high-load scenarios, and maintain high prediction stability even under random load switching and extreme temperature interference.

7.2 Weaknesses

1. **Simplification of High-Frequency Transient Response**
   To reduce computational complexity, the model adopts a first-order RC circuit to approximate the battery impedance characteristics, ignoring the high-order effects of the electrochemical double layer, which leads to a slight lack of accuracy in transient voltage simulation under millisecond-level ultra-high frequency load fluctuations.

2. **Idealized Assumption of Spatial Thermal Distribution**
   The thermodynamic module assumes a uniform temperature distribution inside the battery cell, and does not consider local hotspots (e.g., the area close to the processor) caused by the compact stacking inside the smartphone, which may underestimate the non-uniform aging rate caused by local overheating.

7.3 Promotion

Our modeling framework exhibits high generality and can be extended to other lithium-ion battery-powered devices:

- **Electric Vehicles (EVs):** Replace P_{screen} in the model with P_{motor} (motor power consumption), where the motor power consumption is correlated with speed and acceleration. The impact of ambient temperature T on EV driving range is highly similar to that on smartphones, and our temperature correction factor $f(T)$ can be directly transferred and applied.
- **Drones:** The primary power consumption item becomes $P_{\text{propulsion}}$ (propulsion power consumption), which is related to wind speed and hovering altitude. Time-to-Empty (TTE) prediction is crucial for drones (to prevent crashes), and our continuous-time integral model is safer than simple voltage threshold alarm systems.
- **Wearable Devices:** Due to the extremely small designed battery capacity C_{design}, the weight of background power consumption P_{bg} increases significantly. The model needs to adjust parameter weights and focus on optimizing the sampling frequency of sensors (e.g., heart rate, GPS).

7.4 Conclusion

This paper addresses the challenges of nonlinear discharge and remaining runtime prediction for smartphone batteries by constructing a physics-informed hybrid dynamic model based on the Thevenin equivalent circuit. By coupling the first-order RC differential equations with component-level power consumption data, the model accurately reproduces the phenomena of voltage cutoff and battery recovery. This study reveals the global bimodal distribution characteristic of remaining battery runtime, and identifies high screen brightness, heavy CPU load, and weak signal environments as the core power consumption drivers through global sensitivity attribution (among which, the screen and CPU contribute more than 40% and 70% of the total power consumption in specific scenarios, respectively). Furthermore, electro-thermal coupling and Peukert effect corrections are introduced to verify the robustness of the model under extreme operating conditions. Based on these findings, a Pareto-optimal management strategy that balances energy efficiency and user experience is proposed, which provides a quantitative tool with both physical interpretability and high accuracy for the full lifecycle management of batteries.

RECOMMENDATIONS

Dear Smartphone User,

Greetings!
Based on in-depth research on smartphone battery discharge mechanisms, we have compiled **practical, concise suggestions** to optimize your battery life and eliminate ”battery anxiety”:

1. **Scene-specific Settings for Efficiency**: When watching videos, **reduce screen brightness** (the screen accounts for **over 40% of total power consumption**) and enable **dark mode** for better results. When gaming, reduce frame rate or image quality (CPU/GPU accounts for **nearly 70% of power consumption**)—this is more effective than adjusting brightness.
2. **Avoid Invisible Power Culprits**: In **weak signal environments** (elevators, basements), switch to airplane mode or connect to **stable Wi-Fi** (weak signals force the network module to work at high frequency). Regularly close **background resident apps** to prevent secret power consumption.
3. **Extend Battery Lifespan**: Avoid high-load functions (gaming, navigation) in **low-temperature environments** (cold outdoor) to prevent sudden shutdown. After 1-2 hours of high-load use, let your phone rest for **10 minutes** to restore battery polarization and improve subsequent life.
4. **Monitor Battery Health**: If battery life declines significantly (aging causes **reduced capacity and increased internal resistance**), replace it with an **original battery** in time. Daily use: avoid **overcharging** (unplug after full charge) and **deep discharge** (charge promptly below 20%).

We also expect operating systems to optimize based on these findings: **intelligently adjust CPU frequency and screen brightness** according to scenarios, and **automatically restrict high-power functions** at low battery, to balance battery life and user experience.

We hope these suggestions help make your phone battery more reliable!
Wish you a pleasant user experience!

Sincerely,
Team #2600070
2026

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Report on Use of AI

1. ChatGPT 5.0
   Query1: "How to mathematically model the effect of temperature on battery internal resistance using the Arrhenius equation? And how does the Peukert effect influence effective capacity under high discharge rates?"

   Guidance & Formulation: The AI provided the standard Arrhenius formula ($R = R_{\text{ref}} \cdot \exp(\dots)$) and the Peukert's Law equation. We verified these formulas with textbooks and integrated them into our differential equations to simulate "Electro-thermal Coupling." We modified the notation to match our defined variables in Section 3.

2. ChatGPT 5.0
   Query2: "What is the formula for Normalized Local Sensitivity Analysis to test model robustness? How to interpret the results if the sensitivity index is high for internal resistance?"

   Concept Explanation: The AI explained the definition of the Normalized Sensitivity Index ($S_p = \frac{\Delta Y/Y}{\Delta P/P}$). We used this definition to write the Python code for calculating the indices. The interpretation of the results (e.g., identifying dominant parameters) was derived by our team based on the generated data.

3. ChatGPT 5.0
   Query3: "Rewrite the following paragraph about model robustness verification to make it more academic, highlighting the non-linear coupling effects."

   Refining: The AI suggested improvements to the sentence structure and vocabulary (e.g., changing "checking if the model works" to "rigorously evaluated the model's reliability"). We reviewed the suggestions and retained the ones that best described our mathematical approach.

4. Using ChatGPT 5.0 to assist with translation and code modification.